Estimating a Generalized Gaussian Source

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July 24, 2025

Reference

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Definition (1)

PDF for a generalized Gaussian:

$$f(x) = \frac{\nu}{2\alpha\Gamma(1/\nu)} \exp\left(-\left(\frac{|x-\mu|}{\alpha}\right)^{\nu}\right)$$

- Mean is well-defined, mode $= \mu$.
- $\nu = 2$: Gaussian, $\nu = 1$: Laplacian, $\nu \to \infty$: uniform.
- Most quantization models treat the weight entries as i.i.d. GGD $\Rightarrow$ reduces to scalar estimation.

Fischer Information Matrix (1)

$$\mathcal{I}_{\mu\mu} = \mathbb{E}\left[\left(\frac{\nu \operatorname{sign}(x-\mu)}{\alpha}\left(\frac{|x-\mu|}{\alpha}\right)^{\nu-1}\right)^2\right] = \frac{\nu^2}{\alpha^2}\frac{\Gamma\left(\frac{2}{\nu}\right)}{\Gamma\left(\frac{1}{\nu}\right)}$$

$$\mathcal{I}_{\alpha\alpha} = \mathbb{E}\left[\left(\frac{1}{\alpha} - \frac{\nu}{\alpha}\left(\frac{|x-\mu|}{\alpha}\right)^{\nu}\right)^2\right]$$

$$\mathcal{I}_{\nu\nu} = \mathbb{E}\left[\left(\frac{1}{\nu} + \frac{\nu}{\alpha}\frac{\Gamma'(1/\nu)}{\Gamma(1/\nu)} - \ln\left(\left(\frac{|x-\mu|}{\alpha}\right)^{\nu}\right)\right)^2\right]$$

ML Estimator (1)

If we consider a set of iid Generalized Gaussian $\mathbf{x} := [x_1, \ldots, x_N]$

$$f(\mathbf{x}; \mu, \alpha, \nu) = \frac{\nu}{2\alpha\Gamma(1/\nu)} e^{-\sum_{i=1}^{N} \left(\frac{|x_i - \mu|}{\alpha} \right)^{\nu}}$$

then the log-likelihood is

$$\begin{aligned} \mathcal{L}(\mu, \alpha, \nu) &= \sum_{i=1}^{N} \ln f(x_i; \mu, \alpha, \nu) \\ &= N \left(\ln \nu - \ln 2\alpha - \ln \Gamma(1/\nu) \right) - \sum_{i=1}^{N} \left(\frac{|x_i - \mu|}{\alpha} \right)^{\nu} \end{aligned}$$

ML Estimator (2)

$$\frac{\partial}{\partial \nu}\mathcal{L}(\nu) = \frac{N}{\nu} + \frac{N\Gamma'\left(\frac{1}{\nu}\right)}{\Gamma\left(\frac{1}{\nu}\right)} - \sum_{n=0}^{N-1}\left(\frac{|x[n]-\mu|}{\alpha}\right)^{\nu} \cdot \ln\left(\frac{|x[n]-\mu|}{\alpha}\right)$$

$$\frac{\partial \mathcal{L}}{\partial \mu} = -\frac{\nu}{\alpha}\sum_i \text{sign}(x_i-\mu)\cdot|x_i-\mu|^{\nu-1} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \alpha} = -\frac{N}{\alpha} + \frac{\nu}{\alpha^{\nu+1}}\sum_i |x_i-\mu|^{\nu} = 0$$

and if we take a closer look on α and exploit the nice trnasform-preserving property of MLE

$$\hat{\alpha} = \left(\frac{1}{N}\sum|x[n]-\mu|^{\nu}\right)^{\frac{1}{\nu}}$$

ML Estimator (3)

(i) $\nu = 1$ (Laplacian)

$$\hat{\alpha} = \left(\frac{1}{N} \sum |x[n] - \mu| \right), \quad \hat{\sigma}^2 = \hat{\alpha}^2 \cdot \frac{\Gamma(3)}{\Gamma(1)} = 2\hat{\alpha}^2$$

(ii) $\nu \to 2$ (Standard Gaussian)

$$\hat{\alpha} = \left(\frac{1}{N} \sum (x[n] - \mu)^2 \right)^{1/2} = \sigma$$

(iii) $\nu \to \infty$ (Uniform $[-\alpha, \alpha]$)

$$\hat{\alpha} = \lim_{\nu \to \infty} \left(\frac{1}{N} \sum |x[n] - \mu|^{\nu} \right)^{1/\nu} = \|X\|_{\infty} = \max \left(|x[n] - \mu| \right)$$

$$\hat{\sigma}^2 = \lim_{\nu \to \infty} \hat{\alpha}^2 \cdot \frac{\Gamma\left(\frac{3}{\nu}\right)}{\Gamma\left(\frac{1}{\nu}\right)} = \frac{1}{3}\hat{\alpha}^2$$

Alternative Expression (1)

expression used by Dytso, $X_p \sim \mathcal{N}_p(\mu, \alpha)$

$$f_{X_p}(x) = \frac{c_p}{\alpha} e^{-\frac{|x-\mu|^p}{2\alpha^p}}, \quad c_p = \frac{p}{2^{\frac{p+1}{p}}\Gamma(1/p)}$$

We can see the equivalence of these two expressions by taking the change of variable $\alpha' = 2\alpha^p$

Outline

1. GGD as a Scaled Mixture of Gaussian
2. Gaussian Mixture (Multiplicative Decomposition)
3. Determinacy of a GGD
4. Characteristic Function of GGD
   - Expression
   - Power Series Expansion
   - Asymptotic Behavior

Expression of $e^{\frac{-|x|^p}{2}}$

Theorem 1 (Properties of $e^{\frac{-|x|^p}{2}}$)

- not positive definite for $p \in (2, \infty)$
- positive definite for $p \in (0, 2]$, and there exists a finite non-negative Borel measure μ_p on $\mathbb{R}^+$such that for $x > 0$

$$\mathrm{e}^{-\frac{x^p}{2}} = \int_0^\infty e^{-\frac{t}{2}x^2} d\mu_p(t) = \mathcal{L}_{\mu_p}(s = \frac{x^2}{2})$$

Some comments

- GGD with $p \leq 2$ is **positive definite**, therefore has the nice Laplace transform expression

$$f_{X_p}(x) = c_p \mathrm{e}^{-\frac{x^p}{2}} = c_p \mathcal{L}_{\mu_p}(s = \frac{x^2}{2})$$

- this expression allow us to derive later the product decomposition.
- but the exact form of the Borel measure is not given. if we have this we can do some nice Baysian on it.

$$f_{X_p} = c_p \mathbb{E}_T \left[e^{-\frac{1}{2}x^2 T} \right] = c_p \mathbb{E}_T \left[\phi_{\mathcal{N}(0,1/T)}(x) \right]$$

with $T \sim \mu_p(t)$ as an RV

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GGD = RV · GGD

Proposition 5 (Multiplicative Decomposition of a GGD)

For any (p, q) such that $2 \geq p \geq q > 0$, each $X_q \sim \mathcal{N}_q(0, 1)$ can be decomposed into

$$X_q \overset{d}{=} V_{p,q} \cdot X_r$$

where $r = \frac{2q}{p}$, and $\overset{d}{=}$ denotes the equality in distribution.
$V_{p,q}$ has the following properties

- $p = 2$: $V_{p,q} = 1$.
- $p < 2$: $V_{p,q}$ is a positive RV with a PDF

$$f_{V_{p,q}}(v) = \frac{1}{2\pi} \frac{\Gamma\left(\frac{p}{2q}\right)}{\Gamma\left(\frac{1}{q}\right)} \int_{\mathbb{R}} v^{-it-1} \frac{2^{\frac{it}{q}} \Gamma\left(\frac{it+1}{q}\right)}{2^{\frac{itp}{2q}} \Gamma\left(\frac{p(it+1)}{2q}\right)} dt, \quad v > 0$$

Outline of the Proof

- identify the probability measure

$$1 = \mathbb{P}(X_q \in \mathbb{R}) = \int_0^\infty \frac{c_q}{c_r} \frac{1}{t^{1/r}} d\mu_p(t) = \int_0^\infty dv(t)$$

- identify the random variable

$$\begin{aligned}
\mathbb{P}\left(X_q \in \mathcal{S}\right) &= \int_{\mathcal{S}} c_q \int_0^\infty e^{-\frac{t}{2}|x|^r} d\mu_p(t) dx \\
&= \int_0^\infty \int_{\mathcal{S}} c_r t^{\frac{1}{r}} e^{-\frac{t}{2}|x|^r} dx \frac{c_q}{c_r} \frac{1}{t^{\frac{1}{r}}} d\mu_p(t) \\
&= \mathbb{E}_T \left[\mathbb{P}\left(\frac{1}{T^{\frac{1}{r}}} X_r \in \mathcal{S} \middle| T \right) \right], \quad T \sim v(t) \\
&= \mathbb{P}\left(V_{p,q} \cdot X_r \in \mathcal{S}\right)
\end{aligned}$$

Gaussian and Laplacian Mixture

The GG random variable X_q is a Gaussian Mixture when $q \leq 2$

$$X_q \overset{d}{=} V_{q,q} \cdot X_2$$

and is a Laplacian Mixture when $q \leq 1$

$$X_q \overset{d}{=} V_{q,2q} \cdot X_1$$

We define some notation for the two special cases $V_{p,q}$

$$\begin{aligned} V_{G,q} &= V_{q,q}, \quad \text{for} \quad q \leq 2 \\ V_{L,q} &= V_{q,2q}, \quad \text{for} \quad q \leq 1 \end{aligned}$$

the Gaussian Mixture give us

$$f_{X_p}(x) = \int_0^\infty \mathcal{N}\left(x \mid 0, v^2\right) dF_{V_{G,q}}(v)$$

Some comments

- relate $d\mu_p(t)$ with $f_{V_{p,q}}$?
- $\forall p \leq 2$ GGD can be seen as a Gaussian Mixture
- This make me think about the Gaussian Mixture Model but they use a discrete sum

$$f_X(x) = \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}\left(x \mid \mu_k, \Sigma_k\right)$$

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Determinacy of a GGD (1)

Proposition 2 (Determinacy of a GGD for a given p)

The GGD is determinate for $p \in [1, \infty)$, indeterminate for $p \in (0, 1)$.

- $p \in (0, 1)$. check the Krein sufficient condition of indeterminacy

$$\int_{-\infty}^{\infty} \frac{-\log(f(x))}{1+x^2} dx < \infty$$

For a GGD this reduce to

$$\int_{0}^{\infty} \frac{x^p}{1+x^2} dx$$

And the term is finite when $p \in (0, 1)$, therefore the distribution is indeterminate.

- $p \in [1, \infty)$: Later we will show that the power series expansion of the characteristic function has a positive radius of convergence.

Determinacy of Gaussian Mixture (1)

Proposition 8 (Determinacy of $V_{G,q}$)

The distribution of $V_{G,q}$ is determinant for $q \geq \frac{2}{5}$.

We start the proof by the another Determinacy sufficient condition.

Definition (Carleman's Condition of Determinacy)

If a positive random variable satisfied

$$\sum_{k=1}^{\infty} \left(\mathbb{E}\left[V_{G,q}^{k} \right] \right)^{-\frac{1}{2k}} = \infty$$

then the variable is determinate.

Determinacy of Gaussian Mixture (2)

Notice we have a nice expression for $\mathbb{E}\left[V_{G,q}^k\right]$

$$\mathbb{E}\left[V_{G,q}^k\right] = \frac{\mathbb{E}\left[|X_q|^k\right]}{\mathbb{E}\left[|X_2|^k\right]} \approx \left(\frac{2}{e}\right)^{\frac{k}{q}-\frac{k}{2}} \frac{2^{\frac{k}{2}}}{q^{\frac{k}{q}}} (k+1)^{(k+1)\left(\frac{1}{q}-\frac{1}{2}\right)}.$$

where we approximate the expression using Stirling's formula and thus

$$\sum_{k=1}^{\infty} \left(\mathbb{E}\left[V_{G,q}^k\right]\right)^{-\frac{1}{2k}} \approx \left(\frac{2}{e}\right)^{\frac{1}{4}-\frac{1}{2q}} \frac{q^{\frac{1}{2q}}}{2^{\frac{1}{4}}} \sum_{k=1}^{\infty} (k+1)^{-\frac{(k+1)}{2k}\left(\frac{1}{q}-\frac{1}{2}\right)}$$

The expression diverges if $\frac{1}{2}\left(\frac{1}{q}-\frac{1}{2}\right) \geq 1$ or $q \geq \frac{2}{5}$.

Determinacy of a GGD (with Product Decomposition)

$$X_q \overset{d}{=} V_{G,q} \cdot X_2$$

for the range of values $q \in \left[\frac{2}{5}, 1\right]$

- product of RVs with determinant distributions
- X_q is indeterminate

Takeaways

- Optimization objective relying solely on moments will be unreliable if $p < 1$
- Distortion measurement need to be considered carefully (MSE)
- Characteristic function, likelihood, entropy-based objectives

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Characteristic function $\phi_p(t)$ (1)

Theorem 2 (The Characteristic Function of $X_p \sim \mathcal{N}_p(0,1)$)

Theorem 2 The characteristic function of $X_p \sim \mathcal{N}_p(0,1)$ is given by

- For any $p > 0$

$$\phi_p(t) = 2c_p \int_0^\infty \cos(tx) e^{-\frac{x^p}{2}} dx, t \in \mathbb{R}$$

- For any $p \in (0, 2]$

$$\phi_p(t) = \mathbb{E}\left[\mathrm{e}^{-\frac{t^2 V_{G,p}^2}{2}} \right], t \in \mathbb{R},$$

Characteristic function $\phi_p(t)$ (2)

- $p > 0$: $e^{-\frac{|x|^p}{2}}$ is even
- $p \in (0, 2]$: since we have the decomposition $X_p = V_{G,p} X_2$

$$\phi_p(t) = \mathbb{E}_{X_p}\left[e^{itX_p}\right] = \mathbb{E}_{V_{G,p},X_2}\left[e^{itV_{G,p}X_2}\right]$$

$$= \mathbb{E}_{V_{G,p}}\left[\mathbb{E}_{X_2}\left[\mathrm{e}^{itV_{G,p}X_2} \mid V_{G,p}\right]\right] = \mathbb{E}_{V_{G,p}}\left[\mathrm{e}^{-\frac{t^2 V_{G,p}^2}{2}}\right]$$

notice this can be seen as $\mathcal{L}_{V_{G,p}^2}(\frac{t^2}{2})$

Characteristic function $\phi_p(t)$ (3)

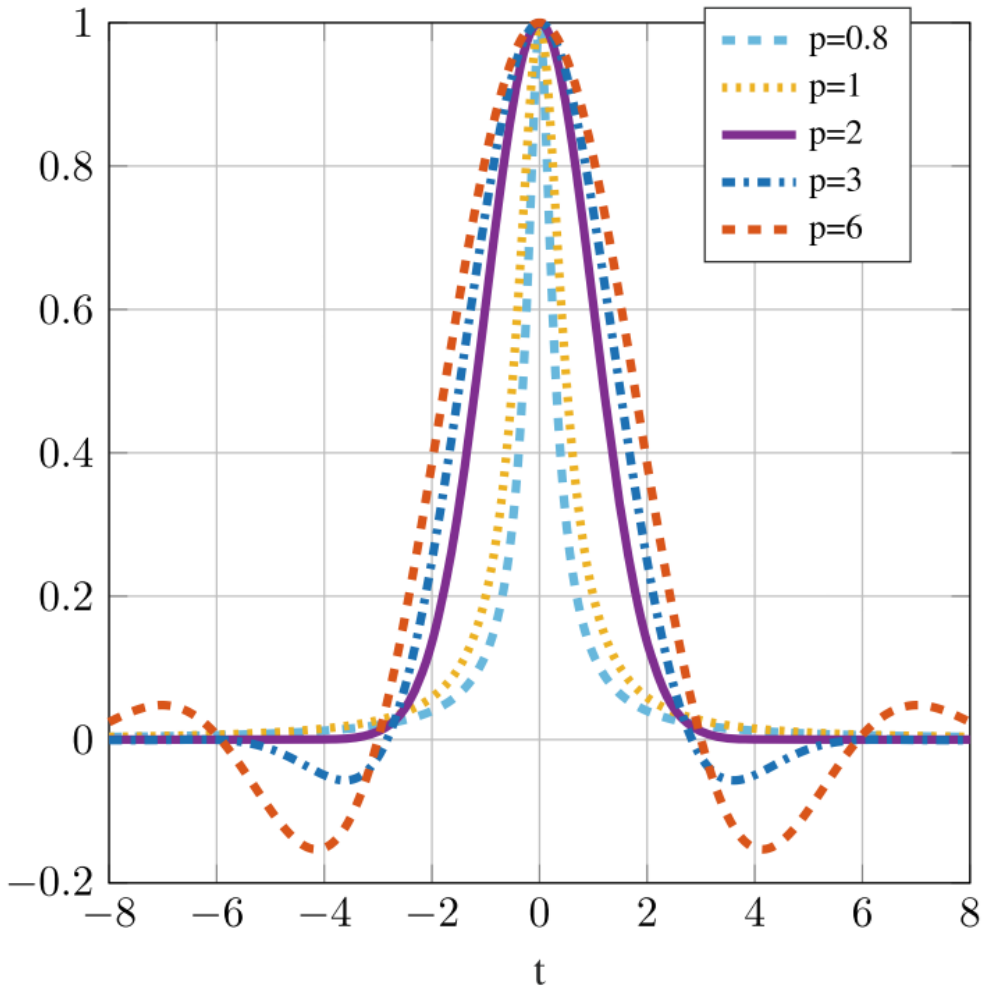


Figure: $\phi_p(t)$ of a $X_p \sim \mathcal{N}_p(0, 2)$ with several different p

Zeroes of the Characteristic Function

Theorem 3 (Distribution of zeroes of the characteristic function)

- $p \in (0, 2]$: $\phi_p(t)$ is a positive function
- $p > 2$: $\phi_p(t)$ has at least one and at most countable zeroes
- $p = \infty$: $\phi_p(t)$ of the uniform distribution has periodic zeroes

Moments of a GGD

The moment pf a GGD can be nicely characterized by the Mellin Transform of the RV.

Proposition 1 (Mellin Transform of $|X_p|$.)

For any $p > 0$ and $X_p \sim \mathcal{N}_p(0, \alpha^p)$

$$\mathbb{E}\left[|X_p|^{s-1}\right] = \frac{2^{\frac{s-1}{p}}}{\Gamma\left(\frac{1}{p}\right)}\alpha^{s-1}\Gamma\left(\frac{s}{p}\right), \mathrm{Re}(s) > 0$$

We can make use of it and look into the power expansion of the characteristic function

$$\phi_p(t) \stackrel{?}{=} \sum_{k=0}^{\infty} \frac{(it)^k}{k!}\mathbb{E}\left[X^k\right]$$

Characteristic function as a power series

Proposition 10

$\phi_p(t)$ is a real analytic function for

- $t \in \mathbb{R}$ for $p > 1$; and
- $|t| < \frac{1}{2}$ for $p = 1$.

For $p < 1$ the function $\phi_p(t)$ is not real analytic.

The proof is followed by taking the ratio test of the power series, exploiting the nice properties given by the ratio of two Gamma functions.

$V_{G,p}^2$ Behavior

Proposition 7 (Asymptotic Behavior of $V_{G,p}^2$)

Let $V_{G,q}^2(v)$ be the pdf of the random variable $V_{G,q}^2$.

$$\lim_{v\to 0^+} f_{V_{G,q}^2}(v) = \begin{cases} 0, & q > 1 \\ \frac{1}{8}, & q = 1 \\ \infty, & q < 1 \end{cases}$$

The proof starts from the power series expansion of $f_{V_{G,q}}(v)$ and $f_{V_{G,q}^2}(\nu) = \frac{1}{2\sqrt{\nu}} f_{V_{G,q}}(\sqrt{\nu})$ ($V_{G,q}$ is a non-negative random variable)

$$f_{V_{G,q}^2}(v) = \frac{1}{2}\frac{\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{1}{q}\right)}\left(a_1 v^{\frac{q}{2}-\frac{1}{2}} + a_2 v^{q-\frac{1}{2}} + a_3 v^{\frac{3q}{2}-\frac{1}{2}} + \ldots\right)$$

$\phi_p(t)$ as $t \to \infty$

Proposition 11(Asymptotic Behavior of $\phi_p(t)$)

With For $m \in \mathbb{R}^+$ we have

$$\lim_{t\to\infty} t^{m+p+1}\mathbb{E}\left[V_{G,p}^m \mathrm{e}^{-\frac{V_{G,p}^2 t^2}{2}}\right] = A_m = \frac{p}{2}\Gamma\left(\frac{m+p+1}{2}\right)2^{\frac{m+2p}{2}-\frac{p+1}{p}}$$

Some thoughts

- **Explicit form for the measure**
  We lack a closed-form expression for the Borel measure μ_p in the mixture representation.
- **Connection with Gaussian Mixture Models (GMM)**
  GGD admits a continuous Gaussian mixture; GMM uses discrete ones. GMM updates with EM method.
- **Indeterminacy for $p < 1$**
  Moment-based optimization becomes unreliable.
- **Characteristic function zeroes**
  For $p > 2$, $\phi_p(t)$ has zeroes. Be cautious when using spectral tools.
- **Asymptotic behavior and heavy tails**
  $\mathbb{E}[V_{G,p}^m e^{-t^2 V_{G,p}^2/2}]$ decays fast for large t, and blows up near $t = 0$ if $p < 1$. This explains sparsity but complicates inference.